Nagizaaz Shaik

LLM & GENERATIVE AI ENGINEER · AGENTIC AI · RAG · LLMOPS

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LLM engineer with five years in data and ML engineering, two shipping production LLM systems (gateways, RAG, agents, fine-tuning) in regulated finance. Doctoral researcher in AI-agent security; 15 merged open-source PRs, including crewAI.

SKILLS

LLMs: Claude, OpenAI, Qwen2.5, Hugging Face, LoRA / PEFT, distillation, INT4 quantization, MLX, vLLM, Ollama
Agentic AI: LangGraph, CrewAI, multi-agent systems, MCP, tool calling, agent memory, Pydantic structured outputs
RAG: LangChain, embeddings, Pinecone, FAISS, BM25 hybrid search, reranking, HyDE, corrective RAG, semantic caching
Evaluation & Safety: RAGAS, LLM-as-judge, A/B testing, OWASP LLM Top 10 red-teaming, prompt injection, Presidio PII
LLMOps & Cloud: FastAPI, Docker, Kubernetes, ArgoCD, GitHub Actions, AWS SageMaker / EKS, GCP Vertex AI, Terraform
ML & Languages: PyTorch, BERT, XGBoost, scikit-learn, reinforcement learning (PPO) | Python, SQL, Go, JavaScript

EXPERIENCE

Doctoral Researcher, Applied AI | Belhaven University, Jackson, MS — Aug 2026 – Present

- Traced a sandbox bypass in an AI-agent code interpreter into 25 open-source projects that copied the code (23 still unpatched); reporter credit on GHSA-m5v3-6ccr-cfgx (working paper *When Vendoring Defeats the Patch*).
- Showed that a planted label flips up to 92% of a LoRA-tuned financial model's predictions across five markets, and that a standard confidence gate misses about half (working paper *Fragile in Different Ways*).

ML Platform Engineer · Contract | Wright-Patt Credit Union ($9B assets), Fairborn, OH — Jul 2025 – Jun 2026

- Built an internal LLM gateway over Claude, OpenAI, and Ollama with evaluation guardrails, cost controls, and RAGAS grounding checks, giving the 12–15-person IT team one approved path to ship generative-AI features.
- Built the credit union's first AWS ML platform as its founding ML platform engineer (SageMaker endpoints, MLflow registry with gated promotion and automated rollback, VPC isolation), cutting deployment failures 85% in 60 days.
- Designed security for all ML workloads (AES-256 at rest and in transit, least-privilege IAM, audit logging) and passed two consecutive NCUA audits with zero findings; codified infrastructure in Terraform with ArgoCD GitOps.
- Mentored two engineers into ML and platform ownership; both now ship production components independently.

Machine Learning Engineer · Contract | Day Air Credit Union ($830M assets), Kettering, OH — Jul 2024 – Jun 2025

- Built RAG document QA over 500+ internal policy and NCUA regulatory documents (LangChain, vector search, Claude), cutting compliance research from about two hours to under 20 minutes per query.
- Fine-tuned a BERT intent classifier (14 classes) and served it with FastAPI at sub-200 ms p99, routing 10,000+ tickets weekly.
- Shipped an XGBoost credit-risk model on SageMaker scoring 500+ loan applications a day, with weekly Airflow retraining gated on AUC, precision, and calibration, and drift monitoring that caught two drift events before they reached credit decisions.

Senior Data Engineer | Cognizant (Financial Services client), Hyderabad, India — Jul 2020 – Jun 2023

- Led an Oracle → Redshift migration, rebuilding 20+ stored procedures as Airflow DAGs and cutting query runtime 60%.
- Built 15 production ETL/ELT pipelines with multi-stage validation that surfaced a silent upstream corruption affecting three months of financial reports; promoted from Associate to Senior in two years.

Data / Python Intern | Uber, Hyderabad, India — Jan 2019 – Oct 2019

PROJECTS

LLM Gateway | Python, FastAPI, Redis, Docker — github.com/shaikn6/llm-gateway

One OpenAI-compatible API over Anthropic, OpenAI, and Ollama with per-tenant caching, rate limiting, A/B routing, cost tracking, and audit logs; SDK contract tests in CI, 415 tests, 99% coverage.

lethe: Agent Memory Server | Go, Python, LangChain, MCP — github.com/shaikn6/lethe

Memory server for AI agents: an RL policy learns from feedback which memories to return, lifting evidence in the top 5 from 0.545 (best hand-set rule) to 0.618 on held-out LoCoMo; 7,119 req/s, crash-safe journal.

Financial Sentiment with LoRA | PyTorch, PEFT, Qwen2.5 — github.com/shaikn6/fin-lora

LoRA-tuned Qwen2.5-0.5B reaches 84.7% on 2,000 news snippets vs 73.6% for FinBERT; a confidence cascade cuts average latency 63% for 0.4 points of accuracy. Adapters published on Hugging Face.

Open Source | 15 merged PRs: crewAI (60K stars, incl. a fix for a bug I reported), Ray (44K), MLflow (28K) — github.com/shaikn6

EDUCATION & CERTIFICATIONS

Belhaven University — Doctor of Business Administration (DBA), Applied AI — Aug 2026 – Present

University of Dayton — M.S. Computer Science · GPA 3.93 / 4.0 — Aug 2023 – May 2025

CMR Engineering College — B.Tech. Computer Science and Engineering — Aug 2016 – May 2020

AWS Certified Machine Learning – Specialty • AWS Certified Data Engineer – Associate • GCP Professional Data Engineer